

DEXTER FERNANDES

Embedded Software Engineer Bristol, UK +44 7436 007369 dexterfernandes11@gmail.com linkedin.com/in/dexterferns
github.com/Dexter-Fernandes

SUMMARY

Software engineer with 3+ years building and shipping real-time C++17 systems on embedded and edge compute hardware, from Intel Movidius VPUs to NVIDIA Jetson platforms. Delivered deterministic real-time inference pipelines to 24/7 industrial field deployments, tripling throughput on embedded hardware and cutting live pipeline crashes by 67% through reliability engineering. Built autonomous robotics systems in ROS2 and C++, including real-time SLAM and multi-sensor fusion on a mobile platform. Practised test-first development with gtest and pytest across the embedded inference stack, and led a four-engineer team through a production two-stage detector deployment in Texas. MSc Robotics, University of Bristol, based in Bristol.

RELEVANT EXPERIENCE

Independent Engineering — Self-Directed Software Engineer

09/2024 – 04/2026 · REMOTE

- Ran a self-directed engineering programme covering modern detection architectures and embedded deployment paths, delivering open-source projects with published benchmarks.
- Rebuilt and benchmarked current detector families (YOLOv8 through YOLO26, RT-DETR, EfficientViT) against production deployment criteria: latency, model size and export path for edge and embedded targets.
- Shipped a real-time multi-stream RTSP video pipeline in C++ and CUDA on constrained edge hardware, published on GitHub.
- Documented a reproducible benchmarking methodology for detector comparison, publishing results and code alongside each project for peer review.

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Hardened deployed services with Dockerised microservices, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Developed a watchdog application that detected and restarted failed services on edge devices autonomously, removing the need for engineer callouts.
- Developed and maintained NVIDIA DeepStream 6.x video inference pipelines for 24/7 industrial perception in harsh desert environments, diagnosing model failures under lighting, motion and environmental stressors and turning findings into fixes for live operation.
- Built a repeatable optimisation workflow from baseline to FP16 to validation, and owned the end-to-end ML lifecycle for real-time perception on BP drilling rigs from data curation through on-site validation, improving F1 by 15% under 24/7 operation.
- Introduced pytest, unittest and linting into CI for edge inference modules, improving release consistency across hardware targets.
- Documented failure modes, recovery procedures and deployment guardrails for the engineers supporting fielded embedded systems.

LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Built asynchronous C++17 perception pipelines on OpenVINO and Intel Movidius VPUs, tripling throughput from 5 to 15 FPS and holding deterministic real-time inference on embedded Intel hardware.
- Drove the C++ inference path test-first, specifying pre-processing, inference and post-processing behaviour as gtest cases before implementation, which made the async refactor from 5 to 15 FPS safe to attempt.
- Profiled C++ inference and pre and post-processing hotspots with Intel VTune Profiler, driving targeted performance fixes in the CNN inference path.
- Deployed quantised vision models using FP16 and INT8 optimisation, structured pruning and layer fusion, halving model size on embedded targets while preserving production accuracy.
- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency and operational cost while holding safety-critical performance targets.
- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets, mentoring junior engineers and coordinating with hardware, software and QA teams to integrate perception modules into deployed systems.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running real-time detection and tracking across 3 concurrent feeds on embedded GPU hardware.
- Implemented a C++ and CUDA TensorRT decode plugin using IPluginV2DynamicExt, moving post-processing into the inference engine to cut latency.
- Converted the detection model to a dynamic-batch TensorRT FP16 engine on 6GB VRAM hardware, profiling p99 latency to sustain 29.7 FPS per stream live.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, ID switches, fragmentation, FPS and VRAM to guide real-time tracker selection.

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR-RGB fusion SLAM, and autonomous navigation.
- Developed ROS2 nodes, launch files and parameterised configurations in C++ and Python, using tf2, message synchronisation and timestamp alignment to debug real-time transform timing issues.
- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, plus an EKF localisation layer with robot_localization fusing wheel odometry and IMU.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations for mapping accuracy and runtime performance.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/Modern-Edge-CV-Benchmarking

03/2026

- Benchmarked YOLOv8, YOLOv10, YOLO26, RT-DETR and EfficientViT on accuracy, latency, model size and deployment suitability for resource-constrained embedded targets.
- Assessed Vision Transformer and foundation-model approaches for detection and segmentation against embedded compute and memory budgets.
- Exported models to ONNX and evaluated OpenVINO deployment paths against edge and embedded hardware constraints.

EDUCATION

University of Bristol — MSc Robotics, Merit 09/2023 – 09/2024 · BRISTOL, UK
Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 (rclcpp) on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, Intel RealSense D435i depth and radiological sensing for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol.
- Configured Ceres, g2o and GTSAM as solver backends and tuned ICP parameters across environments.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a PCL point cloud pipeline.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering 07/2016 – 05/2020 · NAGPUR, INDIA
▪ Core coursework in electronics, circuit design and control systems.

SKILLS

Languages	C++17, CUDA C++, Python, Bash
Embedded and Real-Time	Deterministic real-time inference, asynchronous pipelines, latency and throughput optimisation, resource-constrained embedded deployment, Intel Movidius VPU, NVIDIA Jetson AGX Xavier
Testing and CI	Test-driven development, gtest, pytest, unittest, linting, CI/CD, regression testing, Linux, Git, CMake
Robotics and Autonomy	ROS2 (Robot Operating System 2, rclcpp, rclpy), Nav2, Gazebo Harmonic, tf2, sensor fusion, robot_localization, LiDAR and IMU integration, autonomous navigation
SLAM and Perception	RTAB-Map, LIO-SAM, Cartographer, GTSAM, Ceres, g2o, ICP and scan matching, PCL, Open3D
Edge Inference	TensorRT (custom CUDA plugins), OpenVINO, NVIDIA DeepStream, GStreamer, FP16 and INT8 quantisation, ONNX
Reliability and Operations	Docker, health checks, watchdog services, structured logging, observability, timeouts and retries

ADDITIONAL EXPERIENCE

GrowthStage — Technical Screening Engineer (Contract) 05/2026 – PRESENT · REMOTE
Assess candidate technical depth across perception, SLAM and deployment, running hands-on robotics and computer vision technical screens, and write and calibrate the evaluation rubrics used across those interviews.